Stratified Tube Coercivity and Finite-Box Certificates for the Affine Bernstein Problem

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Abstract

We study the proof interfaces in the September 24, 2026 manuscript on the affine Bernstein theorem in dimensions three through nine. The main local deduction retains the free coefficient in its support-function energy identity. For a tube with k model directions in hypersurface dimension n, we prove an explicit logarithmic Poincare inequality whenever $2k > n + 1 - \sqrt{(n^2-2)/2}$. This includes the strata $k \geq 3$ in dimensions ten through sixteen, but does not eliminate the two-direction obstruction and therefore proves no higher-dimensional Bernstein theorem. Exact completion of squares gives constants in the original Hessian-weighted covector norm. We supply rational examples, an obstruction at $(n, k) = (10, 2)$, an explicit finite-box contradiction scale, and compact-test remainder certificates linked to original Ram local-moment and weak-operator proofs. The source's affine-limit geometry, global section balance, and external interior regularity remain distinguished from these local results. Contribution assessment: C3/5, a proved dimension-and-direction extension of the source tube estimate with explicit physical-norm constants and a precise endpoint obstruction, without an independent global Bernstein proof.

1 Problem, Norms, and Logical Scope

Let $\Omega \subset \mathbb{R}^n$ be nonempty, open, and convex, and let $u \in C^\infty(\Omega)$ have positive-definite Hessian. Write

$$U = \operatorname{cof} D^2u, \quad w = (\det D^2u)^{-(n+1)/(n+2)}, \quad U^{ij}w_{ij} = 0. \tag{1}$$

This is the classical affine maximal equation. Its variational functional is $\int_\Omega (\det D^2u)^{1/(n+2)}\,\mathrm{d}x$, with compactly supported variations. The cached source [1] claims that induced Euclidean completeness forces $\Omega = \mathbb{R}^n$ and u to be a positive-definite quadratic polynomial when $3 \leq n \leq 9$. The induced metric $\delta_{ij} + u_iu_j$ is not the affine metric $(\det D^2u)^{-1/(n+2)}u_{ij}$, and neither is silently identified with the auxiliary tube metric below.

The source also derives an affine-complete immersed-hypersurface consequence using the global graph and completeness theorems of Trudinger–Wang. Those theorems and the interior estimates used at the end are external inputs, not consequences of the finite tests in this report. Reading the full cached scientific text is not an independent verification of every global step. In particular, publication in a public repository is not certification of the claimed new global theorem.

Our contribution concerns the precise energy identity in `sections/tubes.tex`, its logarithmic-cutoff transfer, and the fixed-box contradiction interface in `sections/bounds.tex`. The original

choice $\lambda = 15/16$ handles all $3 \le n \le 9$, $k \ge 2$. Retaining λ yields more than a change of basis: it gives an exact admissibility region for this coercive combination, additional admissible strata, and numerical constants specified by rational algebra. No claim of historical priority is made for completion of squares, Schur complements, weighted integration by parts, or Taylor remainders.

2 The Actual Support-Function System

Fix integers $n \ge 3$, $2 \le k \le n$, and $m = n + 1 - k \ge 1$. On a positive open box $B_0 \subset (0,\infty)^k$, consider a smooth convex stationary tube whose transverse fibers contain zero in their interiors. Let $h(s,e) > 0$ be their support function, $e \in S^{m-1}$. Require

$$B = -D_s^2 h > 0, \qquad R = \nabla_S^2 h + h\mathrm{I} > 0. \tag{2}$$

The support parametrization is $X = (s, he + \nabla_S h)$, with inward conormal $(D_s h, -e)$. Its conormal second fundamental form is $H = \operatorname{diag}(B, R)$. Here and below angular matrices act on $T_e S^{m-1}$. If $m = 1$, the sphere is S^0, angular derivatives are absent, its measure is counting measure, and the determinant of its empty matrix is one.

Put $\delta = 1/(n+2)$ and define

$$\begin{aligned} I_0 &= (\det B)^{\delta} (\det R)^{1-\delta}, & \mathrm{d}A &= I_0 \,\mathrm{d}s\,\mathrm{d}\omega, \\ G &= H/h, & \mathrm{d}\mu &= h^{-n/2}\sqrt{\det B \det R}\,\mathrm{d}s\,\mathrm{d}\omega, \\ f &= \log\left((\det B/\det R)^{\delta}/h\right), & c &= (n+2)/2. \end{aligned} \tag{3}$$

All unmarked base covector inner products below are those of hB^{-1}; angular covectors use hR^{-1}. Set $p = D_s \log h$, $d = D_s f$, and $d_e = \nabla_S f$. In particular, $|D_s v|^2 = h(D_s v)^{\mathsf{T}} B^{-1} D_s v$. This is generally far from the Euclidean coefficient norm.

Every test v is real and in $C_c^\infty(B_0)$, independent of e; all integrals include the sphere. Positivity and smoothness on the compact support justify the integrations below without any completeness of G. The source calculations give $\mathrm{d}A/\,\mathrm{d}\mu = e^{-nf/2}$ and

$$\mu_0 h B^{-1} = (\operatorname{cof} B) h^{-n} e^{-cf}, \qquad \mu_0 h R^{-1} = (\operatorname{cof} R) h^2 e^{cf}. \tag{4}$$

Hessian cofactors in the base and Codazzi cofactors on the sphere have zero divergence. Differentiating the other factors in (4) is essential: freezing them would erase the physical drift.

Lemma 2.1 (Exact weak identities)**.** *Under the preceding stationary-tube assumptions, for every such v,*

$$\int v^2\{k + (n+1)|p|^2 + c\langle p, d\rangle\}\,\mathrm{d}\mu = 2\int v\langle D_s v, p\rangle\,\mathrm{d}\mu, \tag{5}$$

$$\frac{n}{2}\int v^2\{|d|^2 + |d_e|^2 - 2|p|^2 - 2\}\,\mathrm{d}\mu = -2\int v\langle D_s v, d\rangle\,\mathrm{d}\mu - 4\int v\langle D_s v, p\rangle\,\mathrm{d}\mu. \tag{6}$$

Consequently, for any real λ,

$$m_\lambda \int v^2\,\mathrm{d}\mu + \int v^2 x^{\mathsf{T}} K_\lambda x\,\mathrm{d}\mu + \lambda\int v^2 |d_e|^2\,\mathrm{d}\mu = \int v\langle D_s v, b_\lambda^{\mathsf{T}} x\rangle\,\mathrm{d}\mu, \tag{7}$$

where $x = (p, d)$, tensor contraction uses the base covector metric, and

$$\begin{aligned} m_\lambda &= k - 2\lambda, \qquad K_\lambda = \begin{pmatrix} n+1-2\lambda & (n+2)/4 \\ (n+2)/4 & \lambda \end{pmatrix}, \\ b_\lambda &= (2 - 8\lambda/n, -4\lambda/n)^{\mathsf{T}}. \end{aligned} \tag{8}$$

Proof. For completeness, the intermediate stationary identity before cancellation is

$$\int v^2 \left\{ \frac{n}{2}(|d|^2 + |d_e|^2) + (n+2)(|p|^2 + \langle p, d\rangle) - (n-2k) \right\} \mathrm{d}\mu = -2 \int v \langle D_s v, d\rangle \, \mathrm{d}\mu. \tag{9}$$

It follows by inserting the support parametrization in the first variation, using the two divergence-free cofactors, and integrating with (4). More explicitly the scalar support equation is

$$n - 2k + hB^{-1} : D_s^2 f - |p|^2 - (n+1)|p+d|^2 + hR^{-1} : \nabla_S^2 f + 2\langle \nabla_S \log h, d_e\rangle + |d_e|^2 = 0.$$

The base drift is $-np - cd$, the angular drift is $2\nabla_S \log h + cd_e$; their substitution gives (9). Also $hB^{-1} : D_s^2 \log h = -k - |p|^2$. Integrating this last equality with its base drift gives (5). Subtracting twice (5) from (9) cancels the mixed term and gives (6). Add $(2\lambda/n)$ times (6) to (5). The coefficients are exactly (8). These steps reproduce the source identities, rather than postulating an arbitrary positive finite matrix in place of the stationary PDE. □

3 Dimension-and-Direction Coercivity

Let $\mathcal{E}(v) = \int |D_s v|^2 \, \mathrm{d}\mu$. For an admissible parameter define

$$L_\lambda = b_\lambda^{\mathsf{T}} K_\lambda^{-1} b_\lambda, \quad W_\lambda = a^{\mathsf{T}} K_\lambda^{-1} a, \quad a = (n, c)^{\mathsf{T}}, \quad A_\lambda = \frac{L_\lambda}{4m_\lambda}. \tag{10}$$

Theorem 3.1 (Complete admissible range for this combination)**.** *There exists $\lambda > 0$ with $m_\lambda > 0$ and $K_\lambda > 0$ if and only if*

$$k > \kappa_n := \frac{n + 1 - \sqrt{(n^2-2)/2}}{2}. \tag{11}$$

Precisely the permitted parameters form

$$\lambda_- < \lambda < \min\{\lambda_+, k/2\}, \qquad \lambda_\pm = \frac{n + 1 \pm \sqrt{(n^2-2)/2}}{4}.$$

For any one of them, on every tube of Section 2,

$$\int v^2 \, \mathrm{d}\mu \le A_\lambda \mathcal{E}(v), \tag{12}$$

$$m_\lambda \int v^2 \, \mathrm{d}\mu + \frac{1}{2} \int v^2 x^{\mathsf{T}} K_\lambda x \, \mathrm{d}\mu + \lambda \int v^2 |d_e|^2 \, \mathrm{d}\mu \le \frac{L_\lambda}{2} \mathcal{E}(v). \tag{13}$$

The range is necessary only for strict positivity of this particular combined identity, not for every possible analytic argument.

Proof. The determinant is

$$D_\lambda = \lambda(n + 1 - 2\lambda) - (n+2)^2/16 = -2(\lambda - \lambda_-)(\lambda - \lambda_+).$$

Its roots are positive for $n \ge 3$. If $\lambda > 0$ and $D_\lambda > 0$, the other diagonal entry is positive too, so Sylvester's criterion applies. Intersecting this open interval with $\lambda < k/2$ proves (11).

In the actual covector Hilbert space, completing a square gives, pointwise,

$$v\langle D_s v, b_\lambda^{\mathsf{T}} x\rangle - v^2 x^{\mathsf{T}} K_\lambda x \le \tfrac{1}{4} L_\lambda |D_s v|^2.$$

For example the omitted nonnegative square is the squared norm of $vK_\lambda^{1/2} x - \frac{1}{2} K_\lambda^{-1/2} b_\lambda D_s v$. Insert this in (7) and drop the angular term to obtain (12). Absorbing only half the quadratic term gives (13). This calculation is fiberwise valid for every SPD B; it introduces no Euclidean ellipticity ratio. □

Theorem 3.2 (Explicit logarithmic inequality)**.** *Put* $\alpha_i = \log s_i$ *and*

$$\mathrm{d}\sigma = \left(1 + \sum_{i=1}^{k} |D_s\alpha_i|^2\right) \mathrm{d}\mu, \qquad C_{n,k,\lambda} = A_\lambda + k(2 + \sqrt{W_\lambda L_\lambda})^2.$$

Under Theorem 3.1,

$$\int v^2 \,\mathrm{d}\sigma \le C_{n,k,\lambda}\mathcal{E}(v). \tag{14}$$

If $\widetilde{v} \in C_c^\infty(\log B_0)$ *and* $v(s) = \widetilde{v}(\log s)$*, then*

$$\int \widetilde{v}^2 \,\mathrm{d}\sigma \le C_{n,k,\lambda} \int |\nabla\widetilde{v}|^2_{\mathbb{R}^k} \,\mathrm{d}\sigma. \tag{15}$$

The right-hand Euclidean norm is on logarithmic test coordinates only. It does not replace the physical norm in (7).

Proof. The divergence formula (4), together with $hB^{-1} : D_s^2\alpha_i = -|D_s\alpha_i|^2$, gives

$$I_i := \int v^2|D_s\alpha_i|^2 \,\mathrm{d}\mu = -\int v^2\langle np + cd, D_s\alpha_i\rangle \,\mathrm{d}\mu + 2\int v\langle D_s v, D_s\alpha_i\rangle \,\mathrm{d}\mu.$$

By the dual quadratic inequality and (13),

$$J := \int v^2|np + cd|^2 \,\mathrm{d}\mu \le W_\lambda \int v^2 x^{\mathsf{T}} K_\lambda x \,\mathrm{d}\mu \le W_\lambda L_\lambda \mathcal{E}(v).$$

Thus $I_i \le (\sqrt{J} + 2\sqrt{\mathcal{E}(v)})\sqrt{I_i}$, including the zero case, which proves (14) after summing and applying (12). Finally $D_s v = \sum_i (\partial_i\widetilde{v}) D_s\alpha_i$, so Cauchy–Schwarz in the physical covector space gives $|D_s v|^2 \le |\nabla\widetilde{v}|^2_{\mathbb{R}^k} \sum_i |D_s\alpha_i|^2$. This proves (15). □

4 Rational Examples and an Endpoint Obstruction

For $(n, k, \lambda) = (10, 3, 5/4)$,

$$K = \begin{pmatrix} 17/2 & 3 \\ 3 & 5/4 \end{pmatrix}, \quad D = 13/8, \quad m_\lambda = 1/2, \quad b = (1, -1/2)^{\mathsf{T}}, \quad L = 51/13, \quad W = 568/13.$$

Consequently $A = 51/26$ and $C = 51/26 + 3(2 + \sqrt{28968}/13)^2$ is one explicit valid constant. It is not asserted to minimize the logarithmic Poincare constant. Since $\sqrt{28968} < 171$, a completely rational upper certificate is $C < 51/26 + 3(197/13)^2 < 692$.

For $k \ge 3$ and $4 \le n \le 16$, existence also follows directly by choosing λ slightly below $3/2$:

$$D_{3/2} = (-n^2 + 20n - 52)/16 > 0, \qquad k - 2\lambda > 0.$$

The rational choice $\lambda = 23/16$ works simultaneously even on the larger range $3 \le n \le 16$, $k \ge 3$: the determinant is a concave quadratic in n, and it equals $7/128$ at both $n = 3$ and $n = 16$. The choice near $3/2$ does not work at $n = 3$, where $D_{3/2} = -1/16$; the rational choice removes this exception. For $k = 2$, (11) holds exactly for the integers $3 \le n \le 9$. For general n, $\kappa_n/n \to (1 - 1/\sqrt{2})/2$; this is a statement about local coercivity, not an asymptotic classification theorem.

At $(n, k) = (10, 2)$, the mass term requires $\lambda < 1$, but $D_\lambda = -2(\lambda - 1)(\lambda - 9/2) < 0$ on $0 < \lambda < 1$. At $\lambda = 1$ the mass term vanishes and K is singular. Its null vector is $(1, -3)$, whereas $b = (6/5, -2/5)$ pairs with that vector to give $12/5 \ne 0$. Thus even a Moore–Penrose substitution for K^{-1} cannot justify the completion-of-squares estimate: the linear functional does not annihilate the null space. This is an exact negative control for the method. It is not a construction of a smooth nonquadratic affine maximal graph.

5 The Fixed-Box Interface

Let $Q_l = [-l, l]^k$ in logarithmic coordinates. The measure σ in this section means its pushforward under $(s, e) \mapsto \log s$; angular fibers have been integrated out. On compact boxes it has a smooth positive density. A Lipschitz cutoff equal to one on Q_l, zero off Q_{l+1}, with Euclidean gradient at most one, is admissible by compactly supported smooth approximation. Write $M_l = \sigma(Q_l)$ and $C = C_{n,k,\lambda}$. Equation (15) then gives

$$M_l \le C(M_{l+1} - M_l), \qquad M_{l+1} \ge qM_l, \qquad q = 1 + 1/C. \tag{16}$$

The equality boundaries have zero measure. This makes the transition-shell budget exact rather than an estimate using all of Q_{l+1}.

Proposition 5.1 (One explicit contradiction box)**.** *Suppose a sequence of stationary tubes is defined on every fixed logarithmic box eventually. Assume constants $c_0, C_b > 0$ and an integer $l_0 \ge 1$, independent of the approximant, with these two properties: eventually $M_{l_0} \ge c_0$, and for every fixed integer $L \ge l_0$, eventually $M_L \le C_b(L+1)^k$. Assume (15) with the same C on those tubes. These assumptions are incompatible. An explicit sufficient box is*

$$\begin{aligned} &a = \log q, \qquad d = \log(C_b/c_0) + a(l_0 + 1), \\ &L = 1 + \left\lceil \max\left\{ l_0 + 1, \frac{2}{a}\left[d + k\log(2k/a) - k\right]_+ \right\} \right\rceil . \end{aligned} \tag{17}$$

Only after this finite L is fixed does one choose a common large approximating index, including a margin for the outer cutoff.

Proof. Iteration gives $M_L \ge c_0 q^{L-l_0}$. For $x > 0$, concavity of log at $2k/a$ gives

$$k\log x \le ax/2 + k\log(2k/a) - k.$$

For $x = L + 1$ from (17) we have $ax/2 > d + k\log(2k/a) - k$, including when the right side is nonpositive. Thus $ax > d + k\log x$, or $c_0 q^{L-l_0} > C_b(L+1)^k$, contradicting the upper bound. All the finitely many lower bounds, cutoffs, and cell normalizations now require only the maximum of finitely many index thresholds. No limiting measure on the entire logarithmic space, or uniform threshold for all L, has been assumed. □

Which inputs this does not supply. In the source, maximal model-direction number supplies uniform centering of every positive fiber. Positive diagonal base changes and linear transverse maps then provide an interior ball and fixed-box cap bounds. The lower mass comes from positive affine cap area. The upper mass comes from the exact cap identity, integrals of Hessian minors, fiber surface area, and Holder interpolation. In particular the source uses

$$\sqrt{\det B \det R}\,\operatorname{tr} B^{-1} = (I_0 \operatorname{tr} B^{-1})^{\theta}(\det B \operatorname{tr} B^{-1})^{1-\theta}, \qquad \theta = \frac{n+2}{2(n+1)}.$$

The second cofactor budget is genuine geometric information, not implied by the two-by-two matrix K_λ. Proposition 5.1 is conditional on these uniform inputs when it is applied to source models. It does not establish them from finitely many observed cells. Our extended tube range excludes an additional stratum only when the same lower and upper mass inputs have separately been justified for that stratum.

Limit-order negative control. Pointwise convergence on each fixed box alone does not supply a single approximating index for every box. For example smooth densities on $\mathbb{R}$ equal to one on $[-j, j]$ and extremely large outside $[-j-1, j+1]$ converge locally to one but have no common global upper mass budget. This example is not a stationary tube; it isolates the false quantifier exchange that the finite-box argument avoids. Similarly, a finite collection of tested cutoffs does not prove (15) for all compactly supported tests.

6 Ram-Compatible Compact Tests, With Remainders

The original local-moment chapter [5] proves the finite Taylor representation by expanding along actual support segments and retaining the integral remainder. Its differentiated formula keeps all coefficient derivatives. The original operator treatment [6] gives the multi-index weak adjoint

$$\mathcal{L}^*\Theta = \sum_\alpha (-1)^{|\alpha|} D^\alpha (k_\alpha \Theta), \qquad \Theta \in C_c^N(\Omega).$$

Both are useful here precisely because hB^{-1} and μ_0 vary. For the base operator $hB^{-1} : D_s^2$, the weak pairing is $\int f\, \partial_{ij}(\mu_0 h (B^{-1})_{ij}\Theta)\, \mathrm{d}s\, \mathrm{d}\omega$. Neither $\partial_i \mu_0$ nor the coefficient derivatives may be discarded. The cofactor structure in (4), rather than an arbitrary finite inverse, is what permits their exact simplification.

Proposition 6.1 (A physical compact-test enclosure)**.** *Fix a compact base set $K \Subset B_0$ and a smooth cutoff $\chi \in C_c^\infty(B_0)$ supported in K. Let f be smooth on a neighborhood of every segment from a fixed center s_0 to K. Let $T_N f$ be its total-degree Taylor polynomial, $N \geq 1$, and suppose actual derivative bounds give*

$$|f - T_N f| \leq \epsilon_0, \qquad |D_s f - D_s T_N f|_{\mathbb{R}^k} \leq \epsilon_1 \quad \text{on } K.$$

Suppose $hB^{-1} \leq \Lambda \mathrm{I}$ on $K \times S^{m-1}$, and the actual weighted masses $\mu(K)$ and $\sigma(K)$ are finite. Define

$$\begin{aligned} e_0 &= \epsilon_0 \|\chi\|_\infty \sqrt{\sigma(K)}, \\ e_1 &= \sqrt{\Lambda \mu(K)}\, (\epsilon_0 \|D_s \chi\|_\infty + \epsilon_1 \|\chi\|_\infty)\,. \end{aligned} \tag{18}$$

Writing $v = \chi f$, $v_N = \chi T_N f$, one has

$$\|v\|_{L^2(\sigma)} \geq \|v_N\|_{L^2(\sigma)} - e_0, \qquad \mathcal{E}(v)^{1/2} \leq \mathcal{E}(v_N)^{1/2} + e_1. \tag{19}$$

Thus if the right lower bound exceeds $\sqrt{C_{n,k,\lambda}}(\mathcal{E}(v_N)^{1/2} + e_1)$, the hypothesized stationary-tube assumptions cannot all hold on that patch.

Proof. The zeroth-order bound follows directly by integration. Product differentiation gives $D_s(v - v_N) = (f - T_N f) D_s \chi + \chi D_s (f - T_N f)$. Use the physical matrix upper bound, integrate, and apply the triangle inequality in each of the two Hilbert spaces. A strict reversed inequality contradicts (14). This is an enclosure for one actual compact test, not a proof of the inequality from a finite bank of tests. □

For example, if $|s - s_0|_1 \leq \rho$ and every derivative of total order $N+1$ is at most M on those segments, Taylor's integral formula gives $\epsilon_0 = M\rho^{N+1}/(N+1)!$ and $\epsilon_1 = \sqrt{k} M \rho^N / N!$. The second estimate applies the degree-$(N-1)$ Taylor formula to each first derivative, not a formal derivative of a mere value-error bound. This is a direct, boundary-preserving use of the private original remainder proofs. Compact tests keep boundary traces zero; allowing support to touch the edge of a tube would require additional trace terms.

Exact finite polynomial data do not remove the continuum. The moment chapter distinguishes polynomial exactness from polynomial closure: variable coefficients may carry a polynomial outside the retained space. The operator report similarly gives mixed-grade banded matrices, not automatic triangular inversion. Here the physical matrices and masses in (18) still need genuine bounds on all of K, and the derivative remainder still needs actual regularity. At a point, choosing $h = 1$ and $B = \operatorname{diag}(\varepsilon, 1, \dots, 1)$ makes the norm of the first coordinate covector equal to $\varepsilon^{-1/2}$. A Euclidean coefficient error of size $\varepsilon^{1/2}$ therefore has physical size one. No dimension-only Euclidean error conversion is available.

7 Source Architecture and Retained Global Obligations

All five scientific modules, the main file, bibliography, figure source, and README were literally read in bounded displays. The following map states what their use in this report does and does not establish.

Module	Source mechanism	Boundary of this report
`geometry`	Completeness gives compact tangent caps; affine limits, maximal direction number, and thin-slice enlargement give centering.	Read and mapped; no independent general compactness/centering theorem claimed.
`area`	Affine area covariance, stationarity, and the cap identity.	Correct signs and compact-support requirements retained as analytic inputs.
`tubes`	Support parametrization, cofactor drifts, exact cancellation, logarithmic inequality.	Identities rederived locally; free-parameter range and explicit constants proved here.
`bounds`	Positive cap mass and uniform polynomial cell budget; one fixed-box contradiction.	Conditional transfer scale made explicit; uniform geometric mass budgets are not finite-test consequences.
`rigidity`	All-section balance, entire domain, normalized modulus, external interior estimates, blowdown.	Imported final route, not extended by the additional admissible tube strata.

The source's model is a full-dimensional local limit of invertible affine images of the epigraph, with an orthant in k distinguished directions, support in their nonnegative halfspaces, and compact caps for their sum. This is more than a local support function satisfying (2). Maximality over all such models and all admissible coordinates is needed for a common fiber-centering constant. The required convergence index may depend on a chosen normalization; finitely many normalizations may be combined, but an infinite simultaneous assertion would need a new proof.

Once the source has excluded every $k \geq 2$, its one-direction conclusion gives $-\rho K_a(t) \subset K_a(t)$ for every tangent section and every base point and height. It derives ray extension and gap doubling with $\beta = 1 + \rho/2$ and $A = (2 + \rho)(1 + \rho)/\rho$. The normalized modulus uses $1/(8A^{N(r)})$ with $N(r) = \max(0, \lceil \log_\beta(R_*/r) \rceil)$. The external interior estimate then controls normalized second and third derivatives uniformly over section heights, yielding the source's $|D^3 u(0)| \leq Ct^{-1/2}$ blowdown. None of this follows from positivity of K_λ for only some strata. In dimension ten, the surviving possible two-direction stratum is exactly why our local extension is not a new global conclusion.

8 Literature, Comparisons, and Research Limits

The October 9, 2026 primary-source refresh checked the author-hosted Trudinger–Wang paper [2], including its introductory scope, Theorem 2.1, and the displayed Theorem 4.2 and Corollary 4.3.

Their statements and dependency discussion were read; the entire external regularity proof was not independently audited. Its higher-dimensional conclusion needs uniform strict convexity. Its dimension-ten counterexample is singular, not a smooth example violating the hypotheses here. For [3], the publisher record was checked; its global completeness and graph results are used only as source-identified external inputs.

The current arXiv record for Sun–Xing–Xu [4] shows version 1, August 26, 2026. Its abstract announces nonquadratic Euclidean-complete solutions for determinant exponent $a \in [-n/(n+1), 0)$. This does not include the classical exponent $-(n+1)/(n+2)$, which is smaller by $1/((n+1)(n+2))$. The abstract and version record, not its full proof, were checked. Accordingly it is neither evidence for the source's claimed Bernstein theorem nor a counterexample to it.

The public repository path history returned latest path commit `adc7f1241b42`, dated October 6, 2026, in the fresh query (the full hash is in the provenance receipt). This is separate from byte identity between the captured base and October 8 trees. The pinned text was not replaced by remote content. A search and a source-history check do not establish community acceptance or absence of other literature.

Increment over the source and cost. The source already supplies the exact cancellation and a successful uniform parameter. Our additional result is the full admissible (n, k) range for that combination, the extended high-direction strata, and explicit physical-norm constants. The constant supplied here is deliberately conservative: it estimates the k logarithmic covectors separately and reserves half the quadratic energy. It can diverge near a determinant-zero or mass-zero boundary. No claim that it is the best tube, global Bernstein, or computational constant is made. The finite-box scale can consequently be large, and it does not provide an effective approximating index from qualitative compactness.

What the Ram comparison contributes. The original local-moment proof ensures that value and derivative errors are kept separate on the actual support segments. TR208 supplies the exact variable-coefficient adjoint and product rule, with explicitly conditional finite closure. Their useful synthesis here is the compact-test enclosure in the Hessian-weighted geometry, not a replacement of a nonlinear affine PDE by an invertible finite matrix. These classical ingredients are not counted as new general theorems. The core C3 assessment rests instead on Theorems 3.1 and 3.2 and their precisely bounded scope.

Advantages and limitations. The analytic constants are independent of unrecorded Euclidean condition numbers; the physical geometry is preserved throughout. The endpoint negative control detects an actual failed absorption rather than an artifact of floating point. Rational examples can be checked exactly. Conversely, the full stationary identity is indispensable, and neither local SPD data nor small residual samples prove it. The additional strata still require the geometric lower/upper mass inputs before they can be excluded. The test certificate requires certified continuum integrals and derivative remainders, not just quadrature values.

Feasible further work. One can minimize $C_{n,k,\lambda}$ on a compact subinterval of the admissible range with certified endpoint margins, or estimate the logarithmic covectors jointly rather than one at a time. Either would improve constants without altering the basic determinant obstruction. A genuinely new route at $(10, 2)$ would need additional differential information, a different multiplier, or a sign constraint on the null direction. A quantitative global theorem would additionally require

effective centering, cap mass, and convergence-index estimates. Those are open tasks for this report, not promises inferred from its finite algebra.

9 Evidence and Conclusion

The public read scope is all 2,222 lines of nine cached text files: `main.tex` 1–235; `area.tex` 1–167; `bounds.tex` 1–414; `geometry.tex` 1–447; `rigidity.tex` 1–316; `tubes.tex` 1–442; `thin-slice.tex` 1–49; `references.bib` 1–135; and `README.md` 1–17. Their original bytes were hashed before literal display and checked again after reading. There are no unread intervals within these nine files. The external articles cited above have the narrower, explicitly stated access scopes.

Fresh private reads are the local-moment chapter [5], lines 1–300 of 446, and TR208 [6], lines 194–378 and 381–508 of 1,019. The latter ranges contain the jet, bandedness, Taylor, and weak-adjoint proofs. Unread intervals are respectively 301–446, and 1–193, 379–380, 509–1019. The two included chapter files named at lines 92 and 161 of the local-moment chapter were not read and are not treated as evidence. No whole-book or whole-corpus proof audit is claimed.

The owned executable checks passed 169 exact cases with 367 assertions, and 27 separately recorded 65-digit numerical cases with 54 assertions. They test the exact coefficient cancellation, admissible rational parameters, physical-metric completion squares, endpoint obstruction, compact-polynomial fixtures, and fixed-box transfer. The latter numerical diagnostics are not interval certificates. These checks supplement the displayed proofs and do not certify the source's global theorem or the existence of stationary tube data.

The source's dimension restriction is visible in a concrete coercivity problem, but it is not merely a rank test: the exact stationary identity, physical covector norm, cutoff domain, and all geometric quantifiers remain necessary. Retaining the parameter resolves that local problem more fully, extends admissible strata, and isolates the remaining obstruction without changing the global conclusion claimed by the original manuscript.

References

[1] OpenAI, *The affine Bernstein theorem in dimensions three through nine*, September 24, 2026, pinned public manuscript. Public source directory.

[2] N. S. Trudinger and X.-J. Wang, *The Bernstein problem for affine maximal hypersurfaces*, Invent. Math. 140 (2000), 399–422. Author-hosted paper, DOI: 10.1007/s002220000059.

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